

Introduction to Kubernetes

What is Kubernetes?

Kubernetes is an **open-source container orchestration platform**. It is the vital technology that handles the scaling, automates deployment and management of containerized applications across a cluster of machines.

Kubernetes and Microservices: A Practical Example

- Let's say you're running a retail store app with three main microservices:
- Product Catalog Service: A container listing all products and inventory
- Shopping Cart Service: A container is managing user shopping carts
- Payment Service: A container processing orders and payments

(Imagine a customer journey: A customer browses products (Product Catalog), adds items to cart (Shopping Cart), and checks out (Payment). How do these containers communicate with each other? If the Payment Service crashes, how will it heal automatically so that the customer never loses their cart because the Shopping Cart Service is still running. This seamless orchestration is exactly what K8s provides: emphasize resilience, automation, and scale.)

So instead of manually deciding which server each container runs on, monitoring them, and replacing failed ones, Kubernetes does it automatically based on rules you define.

Why Kubernetes Matters in Modern DevOps

Kubernetes has become the industry standard for container orchestration across cloud platforms (AWS, Azure, GCP, on-prem). Here's why it matters for your DevOps career:

- It works on AWS EKS, Azure AKS, Google GKE, and on-premises you learn once, work anywhere.
- Enables high availability and zero-downtime deployments for production systems
- This is the core skill for DevOps, SRE's, Platform Engineers, and Cloud Architects
- K8's is essential for building modern CI/CD pipelines with ArgoCD, Helm, and other GitOps tools

In modern DevOps, Kubernetes isn't optional, it's the foundation. Whether you're optimizing costs, building reliable systems, or preparing for senior roles, Kubernetes proficiency is non-negotiable.

It defines the operation reality of cloud development K8s sets the common language, and API for defining how infrastructure should look, enabling automated, repeatable, and scalable

operations. It runs the declarative infrastructure. With this k8s skill you will understand the core challenge and solutions for things like resiliency, deployment automation and massive scalability.

How mastering K8s can boost your career (roles in SRE, DevOps, Cloud, Platform Engineering)

In the rapidly evolving landscape of 2026, mastering Kubernetes is arguably the highest-leverage moves you can make for your tech career. Why because it sits at the core of how modern organizations build, scale and operate software across DevOps, SRE, Cloud, and Platform Engineering teams. Organizations standardize on K8s for microservices and ML/agent apps, the engineers who can design and operate clusters remain in sustained demand into 2026 and beyond. K8s provides a path to senior roles in Cloud Architecture, SRE, and Platform Engineering.

As a DevOps Engineer mastering K8s helps you to understand how to optimize CI/CD Pipelines, implementing advanced deployment strategies using tools like ArgoCD and FluxCD these tools maintain the desired state by constantly monitoring the repo. You define the infrastructure declaratively, achieving extreme velocity in getting code to production.

In this course you'll work on real-world Kubernetes activities such as cluster lifecycle management, security/RBAC, and networking.